# THE SHEAF OF DIFFERENTIALS PULLS BACK ISOMORPHICALLY ALONG ETALE MAPS OF SCHEMES AND BASES

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2005-2018 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2020-2029, the complete author proof (8 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cotangent triangle lemma and its smooth case (Morphisms, triangle-differentials and triangle-differentials-smooth) and the characterization of etale maps as smooth with vanishing differentials; cited inputs with the two-step argument written in the retained span.

*Source reference.* Original lemma 01UV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2029–2029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01UX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2029–2029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02K4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2029–2029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let*

$$\begin{array}{ccc} X_1 & \xleftarrow{f} & X_2 \\ \downarrow & & \downarrow \\ S_1 & \longleftarrow & S_2 \end{array}$$

*be a commutative diagram of schemes with $X_2 \to X_1$ and $S_2 \to S_1$ étale. Then the map $c_f : f^*\Omega_{X_1/S_1} \to \Omega_{X_2/S_2}$ of Morphisms, Lemma 01UV is an isomorphism.*

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* We recall that an étale morphism $U \to V$ is a smooth morphism with $\Omega_{U/V} = 0$. Using this we see that Morphisms, Lemma 01UX implies $\Omega_{X_2/S_2} = \Omega_{X_2/S_1}$ and Morphisms, Lemma 02K4 implies that the map $f^*\Omega_{X_1/S_1} \to \Omega_{X_2/S_1}$ (for the morphism $f$ seen as a morphism over $S_1$) is an isomorphism. Hence the lemma follows. $\square$